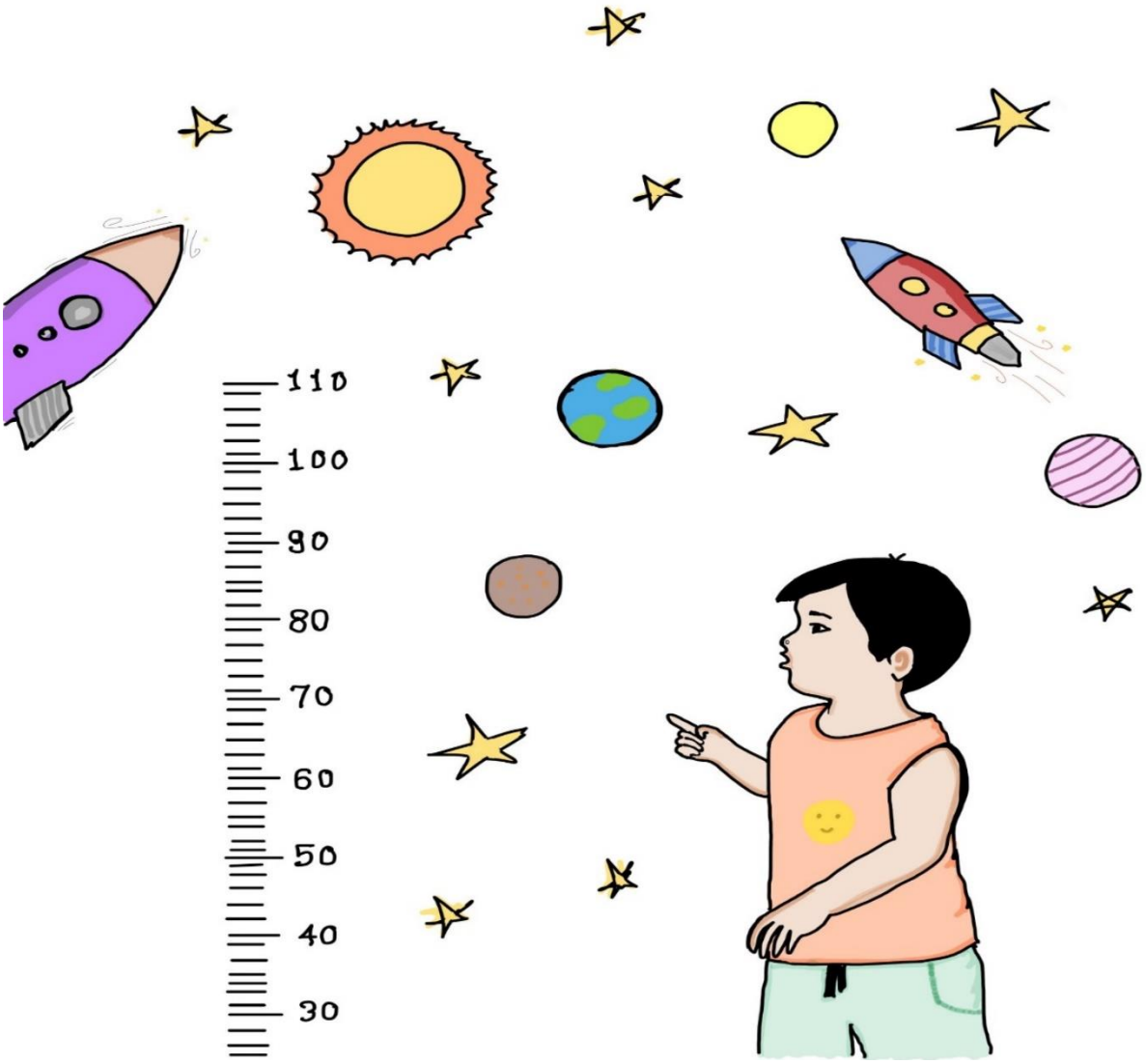
GROWTH & NUTRITION



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GROWTH

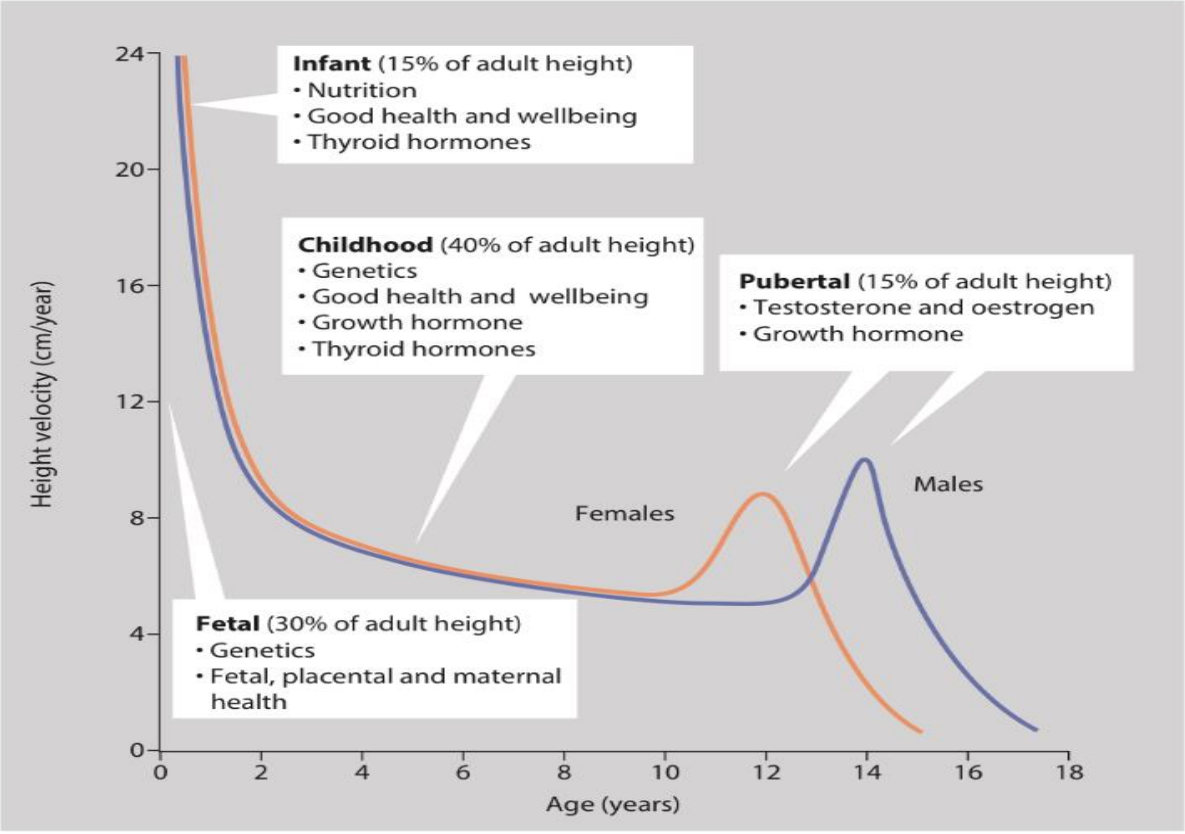
Growth: Is increase in size

Is the most sensitive indicator for health

The growth of all children should be monitored regularly, especially during the first months of life, and recorded on the growth charts .

Growth Phases:

Fetal Phase	Infantile Phase	Childhood Phase	Pubertal Phase
Fastest period of growth ≃30% of eventual height.	Rapid but short & decelerating growth rate ≃15% of eventual height.	Slow, steady but prolonged period of growth ≃40% of eventual height.	Rapid growth (growth spurt) ≃15% of eventual height.
Depends on: 1. Size of mother 2. Placental nutrient supply	Depends on: 1. Good Nutrition 2. Good health 3. Normal Thyroid function–GH . 4. Genetic factors (imp for size)	Depends on: 1. Good Nutrition 2. Good health 3. Growth hormone & T3	Depends on: 1. Sex hormones 2. GH
Problem Sever IVGR or extreme PT –– SS	**Problem** Inadequate rate of weight gain –– FTT	**Problem** Profound chronic unhappiness & decrease GH secretion	**Problem** Early puberty & early fusion of epiphysis



Assessment of growth

ANTHROPOMETRIC MEASUREMENTS

Growth parameters:1. Occipitofrontal circumference(OFC). 2. Weight. 3. Height

1–Weight:–

Physiologic weight loss: Initial weight loss in 1st 3–4 days ≃ 10% in full term, due to (poor suckling,Passage of meconium & diluted urine) ,this weight is regained on the 10th day.







Hanging scale



Stadiometer

Normal birth wt (2.5–4.5 kg)

- Double weight at 4–6 months
- Triple weight at 12 months
- Quadruple weight at 2 years

Rate of wt gain in the 1st year :–

1st 4 months → ¾ kg/mo
2nd 4 months → ½ kg/mo
3rd 4 months → ¼ kg/mo

1–6 years: Wt = (Age yrs x 2) + 8

Length or Height

Height or Length:

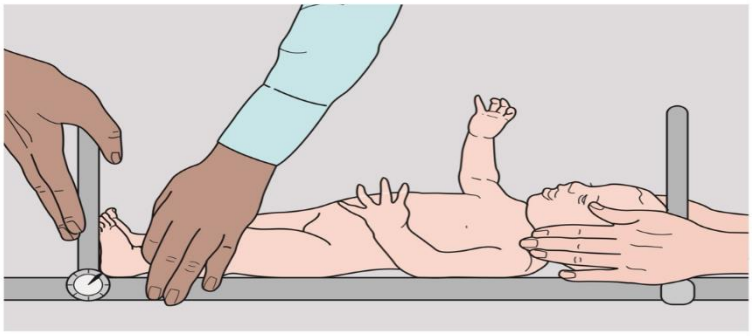
If < 2 years —Supine Length [infantometer]

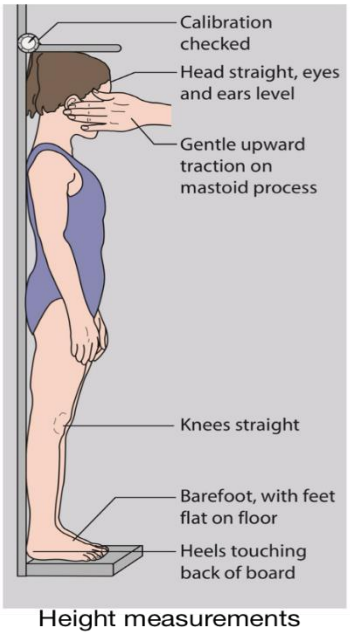
If > 2 years——Standing height [stadiometer]

Length at birth :50cm
At 1 year. :75cm
At 4 years. :100cm

Length >2 years = (Age in years x 5) + 80

Length measurements





Height measurements

OFC (occipitofrontal circumference)

Accurate measurement of head circumference. its value: –

1. Assessment of brain growth

2. Diagnosis, monitoring of small and large head

3– Monitoring of treatment as in hydrocephalus.

- Normal OFC at birth 35 ± 2 cm
- Rate of increase in the 1st Year of life : [1st 2m : 4 cm — 2nd 2m : 3 cm—3rd 2m : 2 cm —— 4th 2m :1 cm ——5th 2m :1 cm ——6th 2m:1 cm]

So it increase ≈12cm in the 1st year, became 47 cm at age of 1year

Abnormalities: microcephaly & macrocephaly,



MID ARM CIRCUMFERENCE (MAC)

Measure: Muscle bulk

Value: Earliest indicator of malnutrition

Best indicator (not affected by SC fat or edema of KWO)

Normal :1–4 years > 12.5 cm Abnormal

Subclinical: (borderline) 11.5–12.5 cm

Severe malnutrition :< 11.5 cm

Method: Using Shakir tap, at mid arm region below deltoid (between acromion & olecranon)





SKIN FOLD THICKNESS

Measure: Body SC fat

Value: Indicator of loss of fat

Best indicator for adiposity (Decreased in Marasmus , Increased in Obesity)

Method: Using Harpenden's or Lang skinfold caliper: At left triceps region or left subscapular region or left supra iliac region

Normal – > 10 mm. – < 6 mm (malnutrition)



Nutritional assessment

Anthropometry
- Weight
- Height
- Mid-upper-arm circumference (MUAC)
- Skinfold thickness

Laboratory
- Low plasma albumin
- Low concentration of specific minerals and vitamins

Food intake
- Dietary recall
- Dietary diary

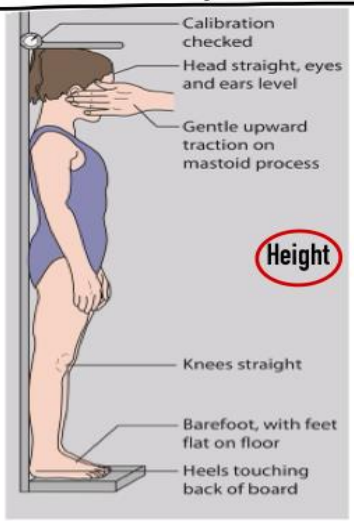
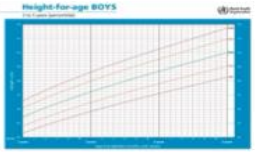
Immunodeficiency
- Low lymphocyte count
- Impaired cell-mediated immunity



Figure 13.12 Mid-upper-arm circumference (MUAC) measurement to identify malnutrition. It is colour-coded; amber is moderate malnutrition, red is severe (<115 mm).

Anthropomorphic measurements

Greeting Introduce self ,Explain to mother what you will do

1-Weight Measurement	1-Adiust the scale to the zero point. 2-Instruct the mother to remove the child's cloth leaving as least as possible of it. 3--Place the child gently on the center of the weight Scale 4-Wait until the scale display stops flashing (digital scale) or the pointer settles (mechanical scale).	
2-Height or length measurement	1-Remove shoes and Place the feet together flat on Height measurement the ground with the heels touching the zero point. 2-Child stands as straight as possible with the heels, buttocks and shoulders touching the measuring device/wall. 3-Be sure the knees are fully extended. 4-Put the head carefully in the neutral position with the lower margins of the orbit in the same horizontal plane as the external auditory meatus.	 Measuring length in infants and young children. An assistant is required to hold the legs straight. Less than 2 years  Measuring height in children. More than 2years
3-Measurement the OFC	1-Prepare a measuring tape 2-Pass the tape on the forehead along the plane midway between the eyebrow and the hairline, to the occipital prominence at the back of the head. 3-Measure 3 times take Maximum د عبدالرحمن الجعكي أخصائي طب الاطفال	 1 Accurate measurement of head
	• Plot the reading on centile chart according to age and sex • Interpretation of chart findings. • Cover the child and thank mother	

Nutritional Requirements

Balanced Diet should contain:

1. Adequate water
2. Adequate calories: CHO = 50% ,Lipids = 35%, Proteins = 15%
3. Adequate macronutrients [CHO, Lipids, Proteins]
4. Adequate micronutrients (Vitamins, minerals, trace elements]
5. Adequate amount of fibers "Fruits, vegetables, cereals" (Bulking effect, prevention of constipation)

Calorie (energy)	water	protein	vit D	vit A	iron
infant :100–110 kcal/kg/day **Toddler** 80–90kcal/kg/day **Preschool**:70–80 kcal/kg/day	rule of 10 1st 10kg *100 2nd 10kg *50 Rest kg *20	infant:2–2.5 g/kg /day	Term infant :400 iu /day Preterm infant:800 –1000 iu/day	2000 iu /day Vit C :50g/day	10–15mg/day

Infants are more vulnerable to poor nutrition because of:

- poor stores of fat and protein
- extra nutritional demands for growth (the weight of a term infant doubles by 4 months and trebles by 1 year of age)
- periods of reduced food intake and increased nutritional demands during illnesses or surgery.

values for energy and protein requirements

Age	Energy (kcal/ kg per 24 h)	Protein (g/ kg per 24 h)
0–6 months	115	2.2
6–12 months	95	2.0
1–3 years	95	1.8
4–6 years	90	1.5
7–10 years	75	1.2
Adolescence	**(male/female)**	
11–14 years	65/55	1.0
15–18 years	60/40	0.8

Malnutrition

Malnutrition: is a pathological state resulting from relative or absolute deficiency or excess of one or more of the essential nutrients.

Worldwide, malnutrition is responsible directly or indirectly for over 30% of all deaths of children under 5 years of age.

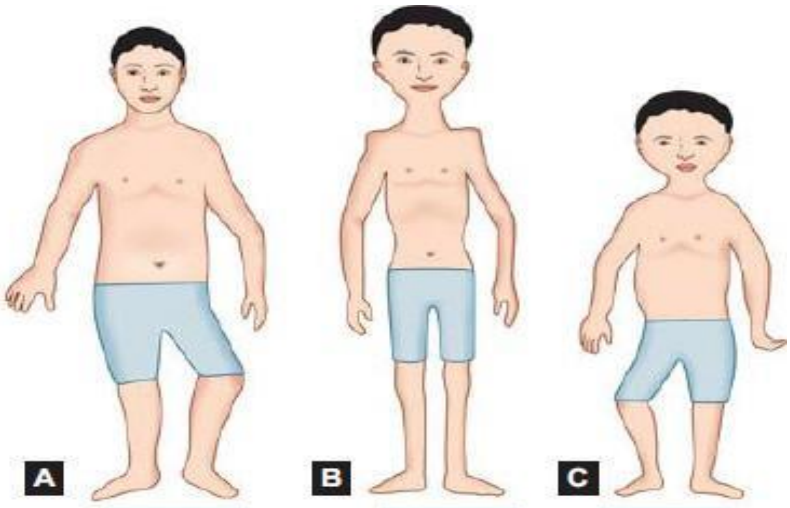
Severe malnutrition is more than 3 standard deviations (SDs) from the median ('z-score -3'), moderate is 2 to 3 SDs below the median.

Types :

1-**Undernutrition**:- state resulting from inadequate quantities of food over an extended period of time,which could be very severe as in marasmus .

2-**Overnutrition**:- state resulting from consumption of excessive quantities of food as in obesity.

WHO CLASSIFICATION OF MALNUTRITION



The best indicator: -

1. Acute malnutrition(**Wasting**) : is **weight for height** (Severe malnutrition is more than 3 standard deviations (SDs) from the median ('z-score -3'), moderate is 2 to 3 SDs below the median)

2. Chronic malnutrition (**Stunting**) : is **height for age** (Stunting is more than 2 SD (more than 2 z-scores) below median)

Welcome classification of "PEM"

% wt for age	Edema	Diagnosis
60 – 80%	–	Simple under nutrition
< 60%	–	Marasmus
60 –80%	+	Kwashiorkor
< 60%	+	Marasmic – Kwashiorkor

other types of malnutrition classification:–

GOMEZ :depends on Weight for Age	WATERLOW:depends onWeight for Height
Mild 1st degree 75–89%	Mild 80–90%
Moderate 2nd degree 60–74%	Moderate 70–80%
Sever 3rd degree < 60%	Sever < 70%

Protein energy malnutrition

Severe malnutrition causes protein–calorie malnutrition, which is classified,depending on the presence or absence of oedema, as:

1. • marasmus
2. • kwashiorkor
3. • a combination ('marasmic kwashiorkor').

	KWO	Marasmus "% more common"
Onset	Acute form of PEM	Chronic form of PEM
Defeicency	Low protein (but enough calorie CHO)	Low calorie (Low protein and Low CHO)
% wt of expected	60 – 80%	< 60%
Cardinal failure	**Edema**(Pitting) Onset :– Dorsum of the feet, Then becomes generalized(LL edema> Dorsum of hands) No Ascites   	**Ms wasting** **Loss of S.C fat :** 1. Abdominal wall& axilla , 2. Buttocks & thighs 3. Face (Senile facies) 
Face	Moon	Senile (old man or wizened appearance)
Hair changes	+++ Sparse, soft & easily detached – Light in color, **Flag sign**: With recovery (Alternating light & dark bands)	± sparse – red –easy preakable

skin changes	+++pigmentation "Flake paint dermatosis",fissures,ulcer ,infections (Site: Buttocks, perineum)  Flaky paint Dermatosis	± ulcer & bed sores Wasted buttocks in marasmus  
Mental changes	+++ (Dull ,apathy–Lethargy – Disinterested in the surroundings – Miserable)	Alert, crying & irritable
Hepatomegaly	+ Fatty liver (reversible)	– –
Appetite	Poor	Good
SC fat	preserved	abscent
Age	(<5yrs) 6m–2yr	(<5yrs) 1st 2 yr
causes	A) **Dietetic** Wrong compensation of scanty breast milk or over–diluted formula by excess CHO Weaning: Substitution of milk with high CHO diet (without proteins) e.g., potatoes, tice water, starch… B) **Infections**:Post–GE: anorexia, vomiting, diarrhea,Post–measles C) **Maternal deprivation**	A)**Dietetic** – Scanty breast milk – Over–diluted formula B)**Infection** (Recurrent GE): : Anorexia & vomiting,Diarrhea C) other chronic diseases as CHD ,malabsorption…ect
Complications	more common	less common

Complication:

1. Dehydration – shock*
2. Hypoglycemia ,hypothermia *
3. Anemia *
4. infection(developing diarrhea, pneumonia and sepsis)*
5. impaired immunity,delayed wound healing.

6. Vitamin deficiency
    - a. Vitamin A: Keratomalacia, xerophthalmia, blindness, bitot spots, infections
    - b. Vitamin B: Stomatitis & glossitis
    - c. Vitamin C: Scurvy
    - d. Vitamin K: Bleeding (Hypoprothrombinemia)
    - e. Vitamin D: Rickets ?? .
7. Heart failure.
8. If sever & prolonged leads to neurodevelopmental delay (impairment in cognitive development)

Laboratory

1. CBC: Anemia, leukocytosis (Infection)
2. Serum albumin: low (N = 3.5–5.5 g/dL)
3. Serum globulins: low
4. Dilutional hyponatremia, Hypokalemia,urea .
5. RBS: Hypoglycemia (Due to disturbed glucose metabolism)
6. Stool & urine analysis–HIV test
7. Serum cortisol,GH high , insulin,IGF low .

Treatment:

- Most children with severe acute malnutrition have an appetite, are alert and can be managed within the community with ready–to–use therapeutic food (RUTF), which has revolutionized its treatment. It is based on peanut butter mixed with dried skimmed milk, vitamins and minerals, and is consumed directly by the child.

- Children with no appetite, severe oedema, a medical complication or are less than 6 months old have complicated severe acute malnutrition and require hospital in– patient care/ it has a high mortality, up to 30%.

Stabilization phase	Rehabilitation phase	Follow up phase
1st week, In ICU Treat life threatening conditions • treat or prevent hypoglycemia urgently • treat or prevent hypothermia • treat or prevent dehydration – but avoid fluid overload. (THE standard WHO ORS contains too much sodium (Na+ 75 mmol/L) and too little potassium for severe acute malnutrition (ReSoMal) (rehydration solution for severely malnourished children). Rehydration should be provided orally, by NGT if necessary. Intravenous fluids are given only for shock • correct electrolyte imbalance – especially potassium and magnesium. Although plasma sodium may be low, they have excess body sodium • treat infection – give broad-spectrum antibiotics/ Treat oral Candida if present • correct micronutrient deficiency – vitamin A and other vitamins/ contained in specialized feeds. Introduction of iron is delayed to second week • initiate feeding small volumes, frequently, including through the night. Too rapid feeding may result in diarrhea & **refereeing** syndrome . Specialized feeds are widely available: initially Formula 75 (75 kcal/100 ml) which is low in protein and sodium and high in carbohydrate is used, subsequently Formula 100 (100 kcal/100 ml) or ready- to-use therapeutic food.	> 2nd week : -Intensive feeding & encourage the child to eat to recover most lost weight (to achieve catch-up growth) -Emotional & physical stimulation are increased -Train the mother to continue feeding at home -Prepare for discharge.	*In home with follow up visits *Continue feeding

Prevention of malnutrition

1. Promotion of breastfeeding: It is the simplest, cheapest & most effective method.

2. General measures: Environmental sanitation, Socio Economic development,family planning ,Adequate food supply, prevention of infection,health education.

3. Nutritional education: Value of breastfeeding, proper weaning...

4. Regular assessment of child health & growth pattern: Growth curves.

5. Regular assessment of nutritional state: Early manifestations of nutritional disorders

6. Promotion of vaccination .

Marasmus Kwashiorkor (MKWO):– is a severe form of PEM.
Definition: Weight for age < 60% with edema (Combined under & malnutrition)
Causes
Mixed form of malnutrition that occurs when

- –Marasmic patient is fed on carbohydrate diet only without protein (common)
- –KWO patient is subjected to carbohydrate restriction (if marked anorexia)

Diagnosis

- –Appearance of edema in a marasmus (but not moon face)
- –Loss of SC fat in abdomen in KW

Failure to thrive(FTT) = Faltering of growth

(It's descriptive term not disease)

Definition: Failure to gain weight in expected way ,or Weight < 3rd centile for age and sex

Also defined as Weight cross 2SD below the mean for age and sex

Or Wt drops 2 major centile within short period

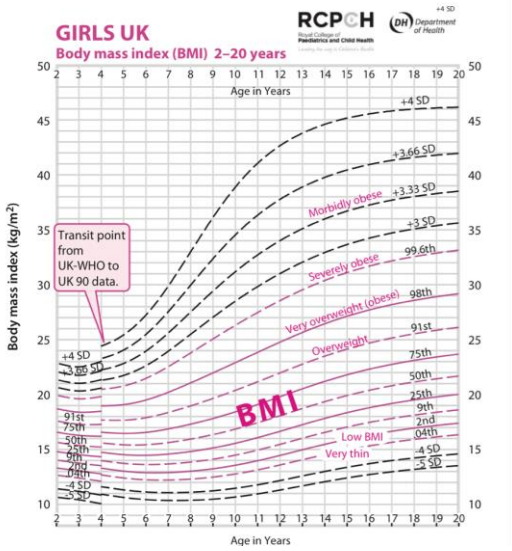
Types & causes :–

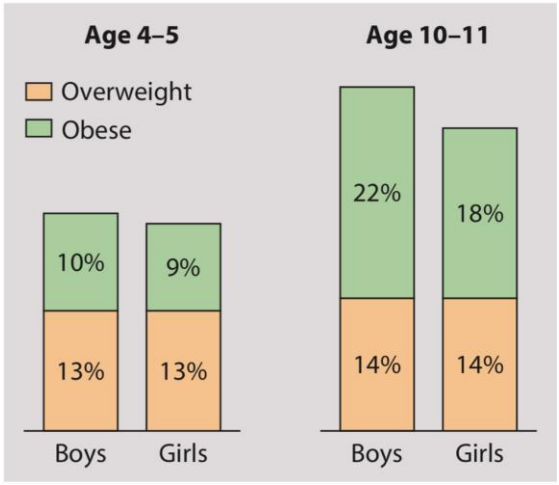
Organic "Pathology" (Less common)	Non Organic "No pathology (more common)
1) Malabsorption As Celiac disease,CF ,CMPI ,zinc deficiency–short bowel syndrome (NEC), biliary atresia	1) Poverty Feeding problems – insufficient breast milk or Insufficient or unsuitable food offered
2) Increase Metabolism (Increase BMR) As hyperthyroidism,chronic infections,HIV ,immune deficiency,malignancy,hypoxia as cyanosis CHD .	2) Neglect
3) Defective utilization As genetic syndrome trisomy,TORCH ,metabolic disease	3)psycho–social deprivation Poor maternal–infant interaction Maternal depression or other mental health disorder Poor maternal cognitive function
4) Increase loss As a severe GERD ,chronic diarrhea.	4) Incorrect preparation as diluted formula poor technique, incorrect preparation of formula
	5) Unstable feeding (habit):Lack of regular feeding times
	6) Impaired suck/swallow Neurological – oro–motor dysfunction, e.g. cerebral palsy Anatomical – cleft palate

Obesity in children

- Is the most common nutritional disorder in the developed world.
- Nowadays % is increasing, obese children are likely to become obese adults.
- The reasons for this marked increase in prevalence are unclear but are due to changes in the nutrition and behavior relating to diet and activity
- Is it important why ? B/C it has short & long term complication
- Wt for age > 97 centile = overweight
- BMI = wt (kg)/ht $(m)^2$ * 100
- BMI > 95 centile for age and sex = obese
- If > 12y BMI ≥ 30

For clinical use, overweight is a BMI over the 91st centile and **obese is a BMI over the 98th centile**. Illustrated textbook"





Types of obesity

	Idiopathic obesity	Obesity (endogenous)
%	Common 90%	5 – 10%
Cause	Multifactorial (familial)	**Endocrine** – as decrease T_3T_4 – DM II,Cushing syndrome, **Syndrome** – as prader willi – turner
Bone Ase	Advanced	delayed usually
Ht	Tall	short
IQ	Normal	Low as in some syndrome
F/H	+++	– ve

Investigation: (Fasting lipid profile–HbA1C– TFT–Chromosomal study–Other I_x according to his examination)

Treatment:

1. **Increase physical activity**:walking or cycling for transport, undertaking household chores, and playing.

2. **Decrease physical inactivity**,limiting television and other small screen recreational activity to less than 2 hours per day.

3. **Healthy diet** (choosing nutrient–rich foods that are lower in energy and glycaemic index)

4.**drugs** (Orlistat is a lipase inhibitor), **surgery** (Bariatric surgery is generally not considered appropriate in children or young people)

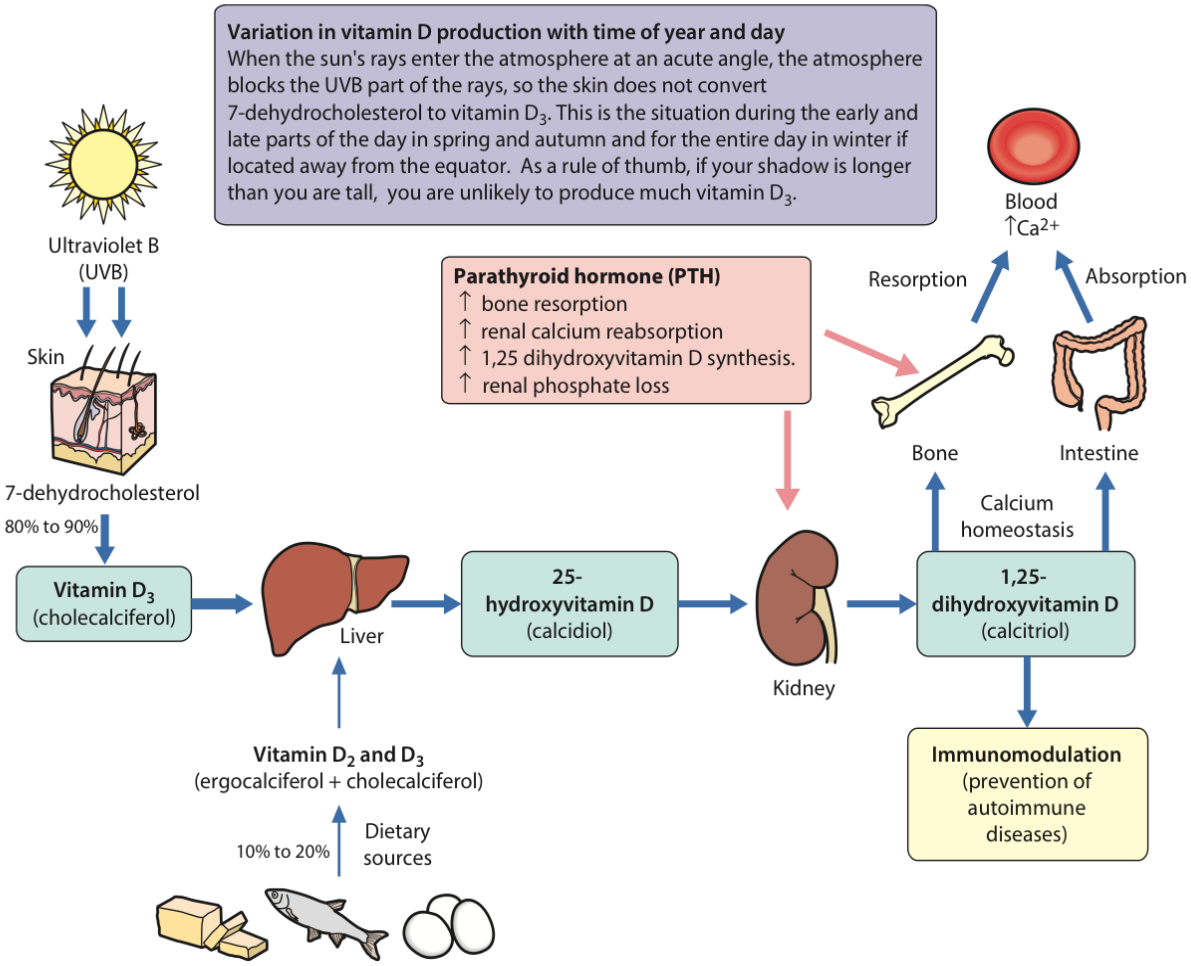
Complication:

System	complications
CVS	Ischemic heart disease –HTN .
Respiratory	Hypoventilation syndrome (daytime somnolence, sleep apnoea, snoring, hypercapnia, heart failure)
CNS	Idiopathic intracranial hypertension
Orthopedic	slipped upper femoral epiphysis, tibia vara (bow legs)
Endocrine	Polycystic ovarian syndrome Type 2 diabetes mellitus,insuline resistance.
Hepatobiliary	Non–alcoholic fatty liver disease Gall bladder disease/gallstone
Psychological	low self–esteem,depression
Others medical conditions:	e.g. asthma, changes in left ventricular mass, increased risk of certain malignancies (endometrial, breast, and colon cancer)

Rickets

- **Definition**: is disease of **growing** bone due to **vitamin D deficiency**.(failure in mineralization of bone tissue)

Vitamin D metabolism:-

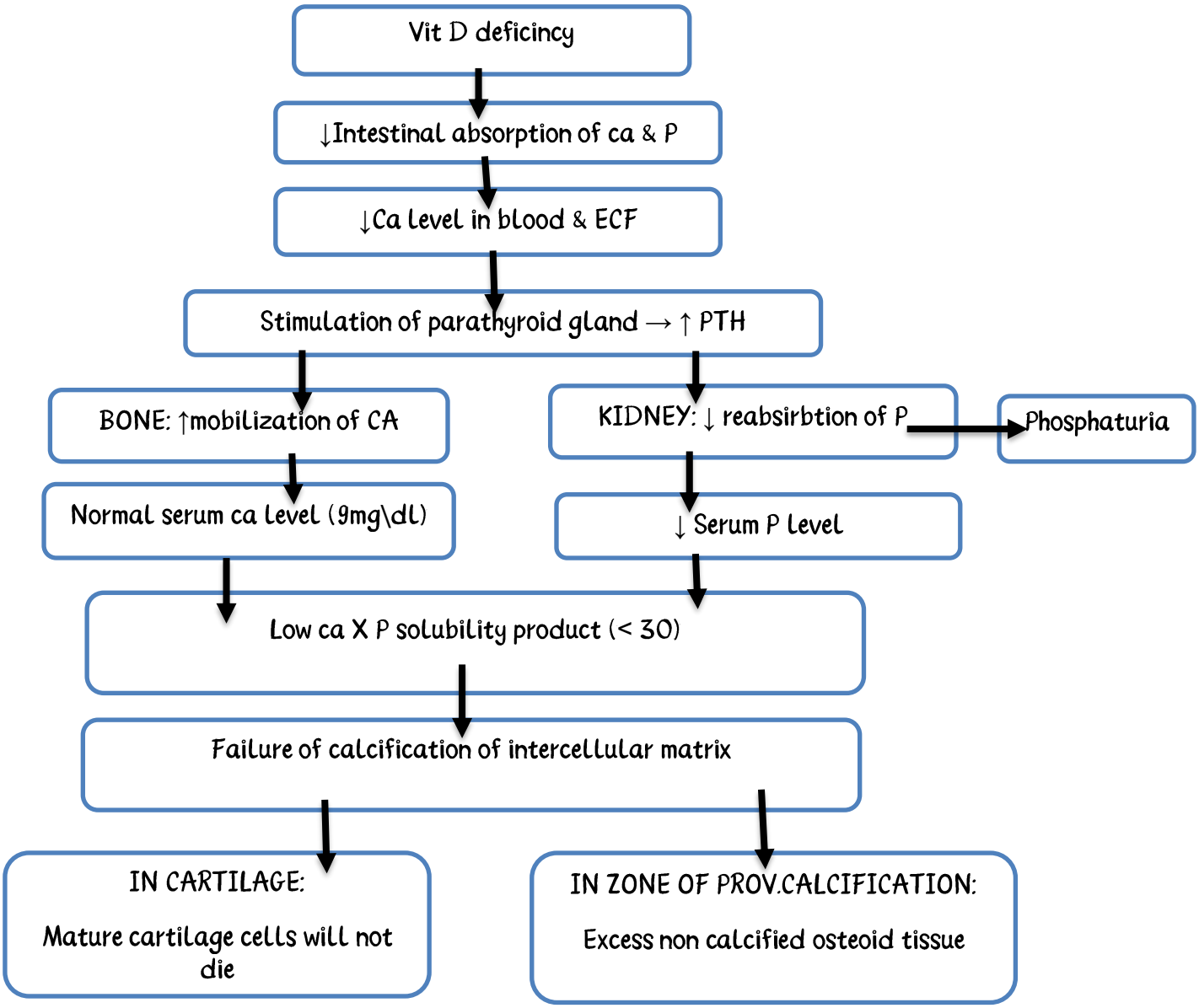


Types of Rickets

Nutritional rickets	Non Nutritional rickets
1. Exclusive prolonged breast feeding > 6m without weaning. 2. Lack of sun exposure: 3W (Wrapping-Winter-time & Windows 3. Diet low in vitamin D or Ca.	1. Hepatic • Biliary atresia • Chronic liver disease 2. Renal • CKD • RTA , Fanconi syndrome • Vit D dependent rickets type I • Familial hypophosphatemic rickets

4. Preterm (osteopenia of prematurity)	**3–others** **Increased breakdown of 25–hydroxyvitamin D** • Enzyme induction by anticonvulsants (e.g. phenobarbital)
5. Other risk factors • Black race . • Malabsorption as CD • Macrobiotic, strict vegan diets • Prolonged parenteral nutrition in infancy	**4–Resistance to effect of 1,25–dihydroxyvitamin D** Vitamin D–dependent rickets type 2

Pathogenesis of infantile rickets:



Clinical Picture

1. Head: Frontal bossing– Macrocephaly (large head)–**Craniotabes 1st sign** – wide & Delayed closure of AF– Delayed dentition & Enamel hypoplasia.

NB: **Craniotabes** is Ping–Pong ball sensation detected on pressing over the occiput (= Bone thinning=soft compressible skull bone), Other important DD :preterm baby–hydrocephalus–osteogenesis imperfecta.

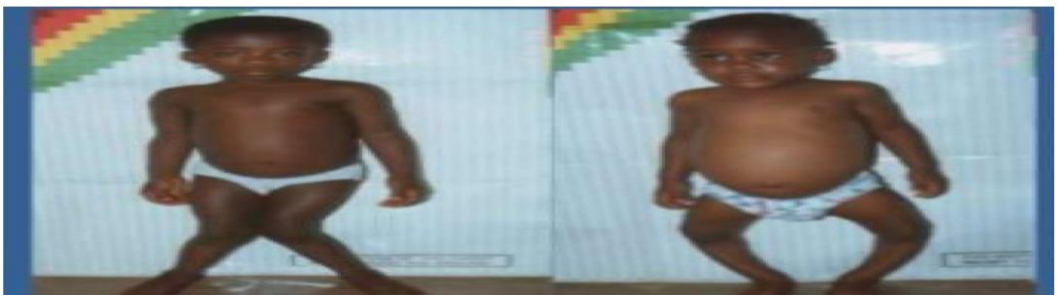




2. Limbs:

- Wide wrist or ankle : Broadening of the ends of long bones(Expansion of metaphyses)
- Deformities of the LL[Genu varum: Bow legs or Genu valgum: Knock knees]
- Deformities of the UL: Convexity of the forearm (in creeping infants)
- Marfan sign: Transverse groove across medial malleolus due to unequal growth of the two ossific centers.





3.Chest

- Rosary beads(Rickety rosary): Cartilage proliferation at the costochondral junctions
- Pigeon chest: increased AP diameter of the chest (Protrusion of the sternum + rib flaring)
- Harrison sulcus: Horizontal groove along the costal insertion of the diaphragm(due to pulling on the soft ribs).





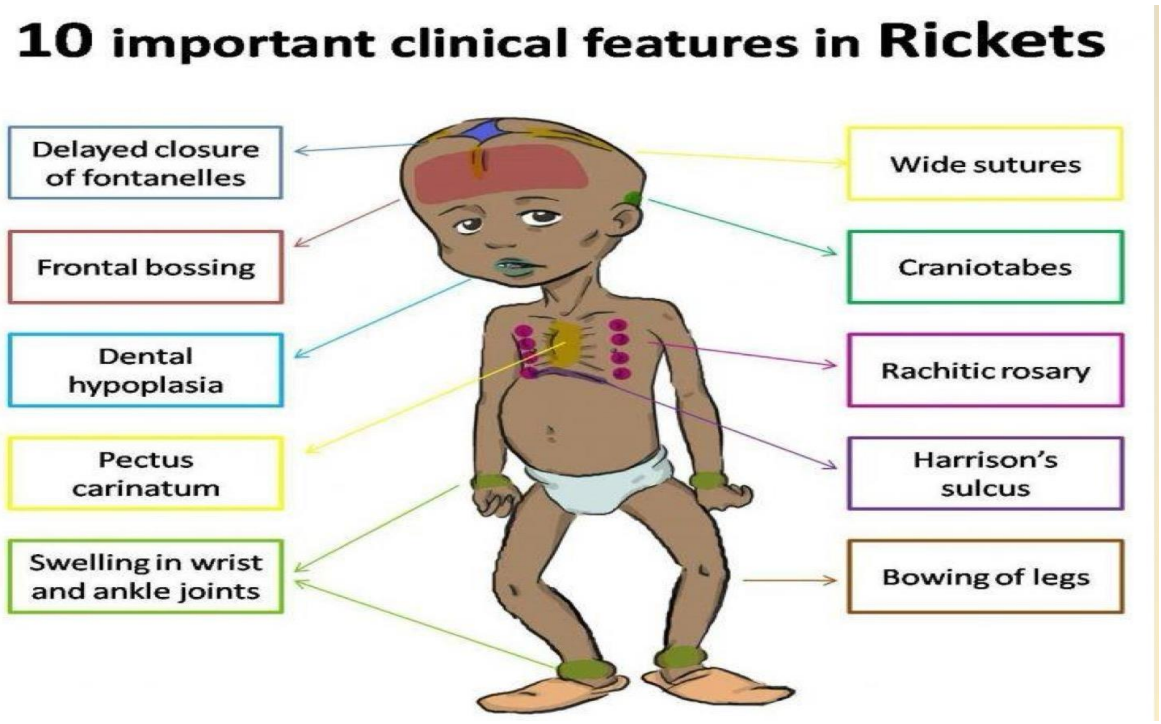
4. Spine: Due to laxity of ligaments(Kyphosis – Scoliosis– Lumbar lordosis)

5.Neurological:

- Hypocalcemic tetany:Carpopedal spasm/ Laryngospasm/ Convulsions.
- Hypotonia.
- Delayed motor milestones

6.Poor growth/short stature,pathological fractures .

7.Cardiomyopathy/heart failure(rare)



Diagnosis:

1. Laboratory tests :-

- serum ca : low (can be normal),normal level : 8.5–10.5 mg/dL
- serum ph : low (high in renal rickets as CKD), normal level: 4.5–6.5 mg/dL
- serum alkaline phosphatase (ALP) : high most sensitive,normal level:50–150 unit/L
- serum PTH : high
- serum vitamin D level : by measuring serum 25–OH vitamin D, which is the most accurate measure of stores of vitamin D in the body.
    - Deficiency: < 20 ng/mL
    - Insufficient: 20–29 ng/mL
    - Normal: 30–100 ng/mL

2. Left wrist & hand x ray

Active rickets: Triad of: widening, cupping, fraying → + wide joint space

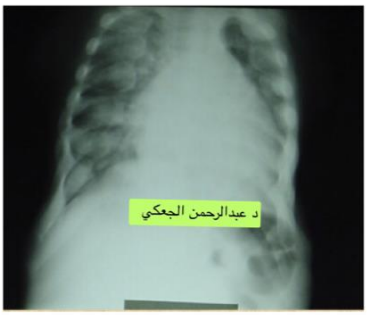
The shaft may show: Rarefaction (↓↓ bone density), Greenstick fracture, Deformity







1. Generalized osteopenia,
2. Coarse trabecular patterns,
3. Frayed paintbrush metaphyseal margins (arrows),
4. Lack of the zone of provisional calcification,
5. Widening of the physis.



Healing rickets

The lower ends of long bone showed:

- **Widening**
- **Cupping**
- **Wide joint space**
- **No fraying**



Healed rickets

- **The lower ends of long bone showed:**
  - **No widening or cupping or fraying**
  - **Normal joint space**
  - **Straight prominent lower end**



Treatment:

- A single high dose (intramuscular) of vitamin D3 (600,000iu),or oral dose 3000–5000 iu /day, followed by the daily maintenance dose. Bone healing starts within 2–4 weeks but may take several years for all bony deformities to resolve.
- If there's hypocalcemia give Ca gluconate (oral or IV if symptomatic)

Prevention:

Advise exposure to sunlight and vitamin D rich diet,start weaning and avoid rachitogenic diet.

- Vitamin D supplement for:
  - For full term: 400 IU/day since birth
  - For premature: 800–1000 IU/day since birth

Complications of Rickets

1. Respiratory: Chest infections, atelectasis .
2. Neurological: Tetany, convulsions, laryngospasm.
3. Short stature (disproportionate)
4. Bone fractures & deformities
5. Iron deficiency anemia as association (Breast milk is deficient in both vitamin D & iron)
6. Contracted pelvis (Obstructed labor later)
7. Complications of treatment (Vitamin D): Hypervitaminosis

Special types of rickets:

	Vit D dependent rickets type I	Vit D dependent rickets type II
Inheritance	Autosomal recessive	Autosomal recessive
Defect	in alpha 1 hydroxylase enzyme	in receptor (end organ resistance)
Onset	3–12 months (early)	3–12 months (early)
C/P	as nutritional rickets	as nutritional rickets
Association	—	short stature +alopecia
25 (OH) D3 level	high	high
1.25 (OH)2 D3 level	low	high
Treatment	1.25(OH)2 D3 and ca supplementation (even parenteral)	high doses of 1.25 (OH)2 D3 and ca supplementation (even parenteral)

Familial hypophosphatemic rickets = VDRR.

X–linked Dominant (severe in boys)

Defect in PCT, leading to:decreased reabsorption of P(lossed in urine) & partial defect in 1 alpha hydroxylation →partial defect activation of vit D

–C/P : usually after 1st year (late) , Dwarfism short stature & marked deformities, minimal involvement of skull & chest, Tetany: rare, non responsive to vit D doses .

Investigation: ser Ca :N or low ,ser Ph :low ,ALP :high –PTH :N , High urinary Ph .

Treatment : oral Phosphate supplementation & 1,25 (OH)2D3.

Hypervitaminosis D

Etiology: Prolonged oral vitamin D therapy

Clinical Picture

- GIT: Anorexia, nausea, vomiting, constipation
- Renal: Polyuria, polydipsia, renal stones (Dysuria, colics, UTI…)

Prevention: Careful vitamin D therapy

Investigations

- Laboratory: high serum Ca, high Urinary Ca
- imaging: X–rays, US (Nephrocalcinosis, stones)

Treatment

Stop vitamin D & Ca therapy, IV fluids, Steroids.

Table: Characteristics of vitamin Deficiencies: mcq

Vitamin	Purpose & source	Deficiency
Water soluble		
Thiamine (B_1)	Coenzyme in keto acid decarboxylation . Occurs in Children from South East Asia with a 'polished rice diet' or those with malnutrition	Beriberi:wet & dry . polyneuropathy, aphonia, encephalopathy, drowsiness, seizures, heart failure, edema,
Riboflavin (B_2)	coenzyme in oxidation reduction reactions	Anorexia, mucositis, anemia, cheilosis, Angular stomatitis, nasolabial seborrhea
Niacin (B_3)	coenzyme in oxidation reduction reactions Malnutrition Regions where maize a major part of the diet	Pellagra: Photosensitivity, dermatitis, dementia, diarrhea, death
Pyridoxine (B_6)	Cofactor in amino acid metabolism	Seizures, microcytic anemia, glossitis, cheilosis, confusion, increased susceptibility to infection neuropathy
Pantothenic acid (B5)	CoA in krebs cycle	nFatigue, apathy, paraesthesia, muscle cramps, hypoglycaemia
Biotin	Cofactor in carboxylase reactions of amino acids	Alopecia, dermatitis, hypotonia, death
B_{12}	Coenzyme for 5 methyltetrahydrofolate formation/ DNA synthesis	Megaloblastic anemia, peripheral neuropathy, posterior lateral spinal column disease, vitiligo
Folate	DNA synthesis	Megaloblastic anemia/ neural tube defects
Ascorbic acid (C) Scurvy مهم جدا	Reducing agent/ collagen metabolism Found in Fresh fruit and vegetables Deficiency is scurvy Rare but can occur in children with very restrictive diets or neurodisability	Scurvy: irritability, periosteal hemorrhage, aching bones purpura, bleeding gums, Petechiae and bruising, gingivitis, coiled hair, poor growth, irritability/ painful joints, impaired wound healing
Fat soluble		
A	Epithelial cell integrity vision [Retinol: Liver, fish liver oils, dairy products Carotenoids: in spinach, carrots, papaya].	Night blindness, xerophthalmia*, Bitot spots, follicular hyperkeratosis/ immune defects **Vit A excess symptoms :-mcq** *Anorexia,Slow growth,Drying and cracking of skin, *Enlargement of liver and spleen *Swelling and pain of long bones, Bone fragility *High ICP(pseudo tumor cerebri) *Alopecia, Carotenemia *Fetal abnormalities

D	Maintain serum calcium, phosphorus levels. [90% from ultraviolet B other :Eggs, Fortified foods margarine, yoghurts .	Rickets: reduced bone mineralization
E	Antioxidant [Vegetable oils]	Hemolysis in preterm infants/ retinopathy, progressive neuropathy areflexia, ataxia, ophthalmoplegia
K	Post translational carboxylation of clotting factors II, VII, IX, X .	Prolonged prothrombin time/ hemorrhage/ haemorrhagic disease of the newborn)

Breast Feeding

- Breastfeeding should start soon after birth & exclusive breastfeeding for the first 6 months is the current World Health Organization (WHO) recommendation,
- **Breast is the best (B/C provides the ideal nutrition for infants during the first 4 months to 6 months of life & its life-saving in developing countries)**

The first breastfeed should take place as soon as possible after birth:-

Colostrum	Transitional milk	Mature milk
1st 5 days	5 – 21 days	≥ 21 days
Small amount	Large	Very large 1-2 L/D
Yellowish and thick	Whiter	Whiter and thinner
Alkaline	–	Neutral
High calorie, protein and IgA and leukocyte & it primes the gut microbiome with gut flora.	Protein & IgA decrease	Fore milk (early, watery and high CHO)
Low fat and CHO	Fat & CHO increase	Hind milk (late, creamy and high fat)

Exclusive breastfeeding in early infancy is life-saving in low and middle-income countries.

Physiology of breastfeeding :-

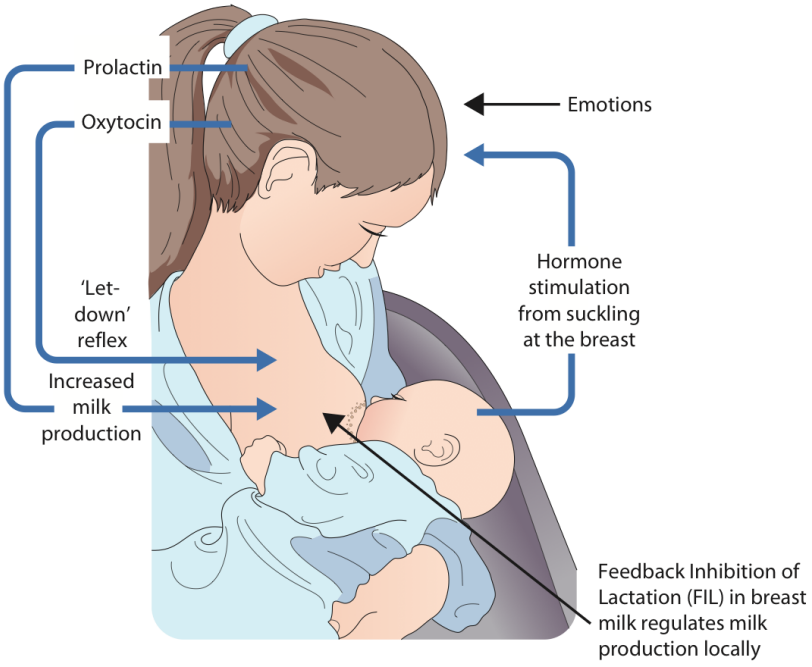
1. **Baby** uses rooting, sucking, and swallowing reflexes to locate nipple and feed

2. **Tactile receptors** in nipple activated

3. **Hypothalamus** sends efferent impulses to anterior and posterior pituitary

4. **Anterior pituitary**
Prolactin secretion stimulates milk secretion by cuboidal cells in the acini of the breast

5. **Posterior pituitary**
Oxytocin secretion results in contraction of myoepithelial cells in the alveoli, forcing milk into larger ducts – the so-called 'let-down' reflex



gure 13.6 Physiology of breastfeeding

Breastfeeding proper position & attachment.

Proper position of baby while breastfeeding includes

1. Supporting whole of baby's body.
2. Ensure baby's head, neck and back are in same plane.
3. Entire baby's body should face mother.
4. Baby's abdomen touches mother's abdomen.

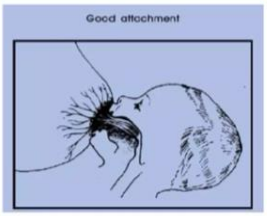
- ***Correct positioning will ensure effective sucking and prevent sore nipples and breast engorgement.***

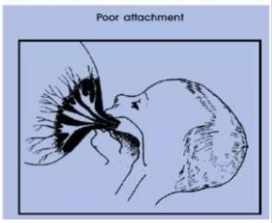


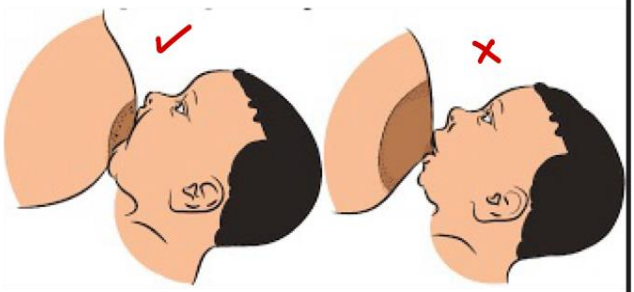
Attachment of baby on mother's breast

- Four signs of good attachment are:

1. Baby's mouth wide open.
2. Lower lip turned outwards.
3. Baby's chin touches mother's breast.
4. Majority of areola inside baby's mouth.







Criteria of Adequacy of feedings :–

- Baby is calm & satisfied after each feed & Sleeps well (2–4 hours)after feeds.
- Normal motions–Normal amount of!urine
- Normal weight gain (most important).

Advantages & disadvantages of breastfeeding:–

Advantages	Disadvantages
I–Advantages for the infants: 1. Immune valve:– (decrease risk of AGE – URTI, OM – Sepsis and NEC in preterm) • secretory **immunoglobulin A (SigA)**, the primary protective agent, which coats the intestinal mucosa and prevents bacteria from entering the cells • human milk **oligosaccharides**, which function as prebiotics, preventing pathogenic bacteria from attaching to mucosal surfaces • **white blood cells**, which can kill microorganisms • whey proteins (**lysozyme** and **lactoferrin**), which can kill bacteria, viruses and fungi. 2. Increase cognitive function(overall increase in IQ (intelligence quotient)) 3. Decrease the risk of diseases (DM type 2 – Obesity–HTN – Atopy – Eczema – CD – SIDS) 4.Enhances the mother–child relationship. 5. Sterile , always available 6. Without cost. 7. Live saving , decrease malnutrition. **II–Advantages for the mother:** 1.Decrease breast and uterine cancer 2.Natural Contraceptive 3.Construction of uterus decrease PP Hge 4.Lowers the risk of developing postnatal depression	1. Transmission of infection as HIV,CMV… 2. Transmission of drugs as chemotherapy or anti thyroid drugs,lithium. 3. Low in :– • Vit D – : > 6m rickets • Vit K – risk of HDN (vit K IM at birth) • Iron – : > 6m IDA 4. Breast milk jaundice(mild, self–limiting, unconjugated hyperbilirubinemia/ advice to continue breastfeeding/ assess infant if prolonged) . 5. Unknown intake. 6. Less flexible. 7.Physical breast problems – sore, cracked nipples, thrush, mastitis, inverted nipples, inadequate milk supply. 8.Emotional upset:If difficulties or lack of success, can be upsetting

Scientific explanation of some of the properties of breast milk

Anti-infective properties

Humoral	
Secretory IgA	Comprises 90% of immunoglobulin in human milk. Provides mucosal protection, but of uncertain benefit
Bifidus factor	Promotes growth of *Lactobacillus bifidus*, which metabolizes lactose to lactic and acetic acids. The resulting low pH may inhibit growth of gastrointestinal pathogens
Lysozyme	Bacteriolytic enzyme
Lactoferrin	Iron-binding protein. Inhibits growth of *Escherichia coli*
Interferon	Antiviral agent
Cellular	
Macrophages	Phagocytic. Synthesize lysozyme, lactoferrin, C3, C4
Lymphocytes	T cells may transfer delayed hypersensitivity responses to infant. B cells synthesize IgA
Nutritional properties	
Protein quality	More easily digested curd (whey-to-casein ratio: 60:40)
Lipid quality	Rich in oleic acid. Improved digestibility and fat absorption
Fat metabolism	Enhanced lipolysis, from breast milk lipase
Calcium-to-phosphorus ratio of 2:1	Reduces hypocalcaemic tetany and promotes calcium absorption
Renal solute load	Low
Iron content	Bioavailable (40% to 50% absorption)
Long-chain polyunsaturated fatty acids	Structural lipids; important in retinal development

Absolute contraindications of breastfeeding

Maternal :–

1. HIV mother in the developed world .
2. Radioactive iodine –chemotherapy.
3. Active HSV at nipple.
4. –Active maternal TB ?!.

Infantile :–

1–infant with Galactosemia .
2–infant with phenylketonuria ?!.

Maternal hepatitis:

If mother HbsAg positive during pregnancy: Allow feeding +
Passive immunoglobulin & HBV vaccine to baby immediate after birth (within 12 hours)
—–
If mother hepatitis C positive during pregnancy or nursing: Allow feeding

Compounds	Human milk	Cows milk
1. **Protein (higher in cows milk)** a. Dietetic protein • Soluble (Lactalbumin) • Insoluble (casein) • Soluble/Insoluble ratio b. Non dietetic protein • Lactoferrin level • Immunoglobulins • Lysozymes level • Essential amino acids	60% 40% 3:2 (protein is fine and thin and easily digested High – static to E.coli and increase iron absorption High (specific to human pathogens) High – bactericidal High – essential for brain development	20% 80% 1:4 (protein is tough and thick and hardly digested Traces Traces Traces Traces
2. **Fat (≃ the same amount)** • Fat globules size • Diurnal variation • Lipase enzyme level • Essential fatty acids	Smaller size – easy digestion Present – high concentration (night) High – help digestion Higher (11%) especially linoleic and oleic acids	Larger size – hard digestion Absent Lower level Lower level
3. **Carbohydrate (higher in breast milk)**	β Lactose – no fermentation (no gasses or vomiting) Some is converted to lactic acid Increase calcium absorption Bacteriostatic effect	α Lactose – high incidence of fermentation excess gasses and vomiting
4. **Minerals & vitamins.** • Amount • Ca:Phosphate ratio • Sodium content • Iron • Vitamins • Calories	Low 2/1/ so absorption is better and rickets is less common Low (less renal solute load) Low with good absorption ≃50% sufficient for 1st 4–6 months Low in vit D&K , but high in vit C & niacin . Each 100ml =67 calories	High 4/3 so less absorption – high risk of rickets High **Very** low with bad absorption ≃10% (less bioavailable) Low in vit C & D but high in thiamine riboflavin . Each 100ml =67 calories
5. Bacterial content	Sterile	Liable to contamination

Artificial (Formula) feeding

Definition:– the infant is given other milk rather than the breast milk during the period when the infant would naturally receive the human milk

Types :–

1–Substitutive feeding (all breast feeds are replaced by bottle feeding)

- Contraindications of breast milk (maternal + infant): as before, especially absolute CI.
- Absent mother (work, travel, death)
- Absent nipple & Absent breast milk (Agalactia)

2–Complementary feeding (In each feed begin with breast & complete with artificial) Due to Scanty breast milk & multiple babies.

Precautions

- Never prescribe milk on mother request.
- Test feed or Test weight to insure that breast milk is insufficient
- Prescribe humanized milk formula.
- Avoid sweet milk as the baby may refuse the mother's breast.
- Breast milk should be given first completely, then feed is completed by artificial Use cup & spoon (no bottle feeding to avoid nipple confusion)

3–Supplementary feeding (some feeds are replaced by bottle):

- Working mother: during working hours only if work > 4 hours
- Twins: alternative feeding one baby & supplementary artificial feed the other.

Types of milk formula – Special type and ordinary formula: Special formula as

1. AR anti reflux formula – GERD
2. HA hypoallergenic or hydrolyzed formula or amino acid based formula (neocate)– CMPI
3. LF lactose free formula – Lactose
4. PT preterm formula – Preterm baby
5. Follow on formula : > 6m .

Amount of milk needed in each infant according to the weight :–

Normal infant usually needs 110 calories /kg /day .

- Each 100ml of milk =67 calories .
- So the amount of milk needed per day =
- The amount of milk /feed = The amount of milk per day / nomber of feed per day .

$$\frac{100 \times 110 \times \text{body wt}}{67}$$

Weaning

Definition :– Its gradual introduction of solid or semisolid food rather than milk.

Onset :–at age 4–6 months [B/C inc infant calorie & vit ,iron needs , mature GIT ,also developmental becomes mature]

- If started < 4m : early weaning (increase of CD)
- If started > 6m : Late weaning (increase of IDA and rickets)

10 Steps of weaning :–

1– Start By Small Amount: Gradually Frequency & Amount

2– Start with Cereals Veg. & Fruits , Meats & Eggs.

4–6m :– Cereals + Corn (Pudding, Biscuits) 7m: + Rice + Fruits 8m :yogurt –vegetable in water . 9m: + Beans + Vegetables in soup 10m :+ Liver + Fish 11m :+ White meat (Chicken & Rabbits) 12m: + Red meat >1year :+ Eggs + fresh Animal milk + Honey (Botulism)

3– One New Food at Time , spaced By 3– 4 days

4– Food Color, Taste & Form (not spicy or sweet) Must Be Attractive

5– Eating with Family Members

6– Avoid Forcing To Eat, re Introduce Disliked Food 2–3 wks Later

7– Meal Time : 15 – 30 min.

8– Parents Should Never Show Dislike Toward Food

9– Avoid Distraction During Meals

10– Strict Hygiene before and After Meals

Differential Diagnosis (Fontanelle abnormalities): mcq

Depressed AF	Pulging AF	Wide AF	Delayed closure AF	Early closure AF
1. Dehydration 2.Hypovolemic shock 3.Malnutrition as marasmus	High ICP 1. Meningitis 2. Encephalitis 3.Hydrocephalus 4. Brain tumor in infant 5. IC Hge as in preterm	Size > 5cm 1.Hydrocephalus 2. Rickets 3. Down S 4. Hypo T_3T_4 5.Achondroplasia 6. Osteogenesis imperfecta	> 18m still open 1. Hydrocephalus 2. Rickets 3. Down S 4. Hypo T_3T_4 5. Achondroplasia 6. Osteogenesis imperfecta	Closure at birth or before 6m of age 1.Hyper T_3T_4 2.Craniosynostosis

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تنبيه:

هذا العمل تم بعد جهد كبير من قبل الدكتور الاخصائي: عبدالرحمن الجعكي، ويعتبر ثمرة سنين من خبرة ممارسة طب الأطفال، كل ذلك كان ممزوج بحنكة العطاء الاكاديمي للمادة

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